Title: Smart Driver Safety System

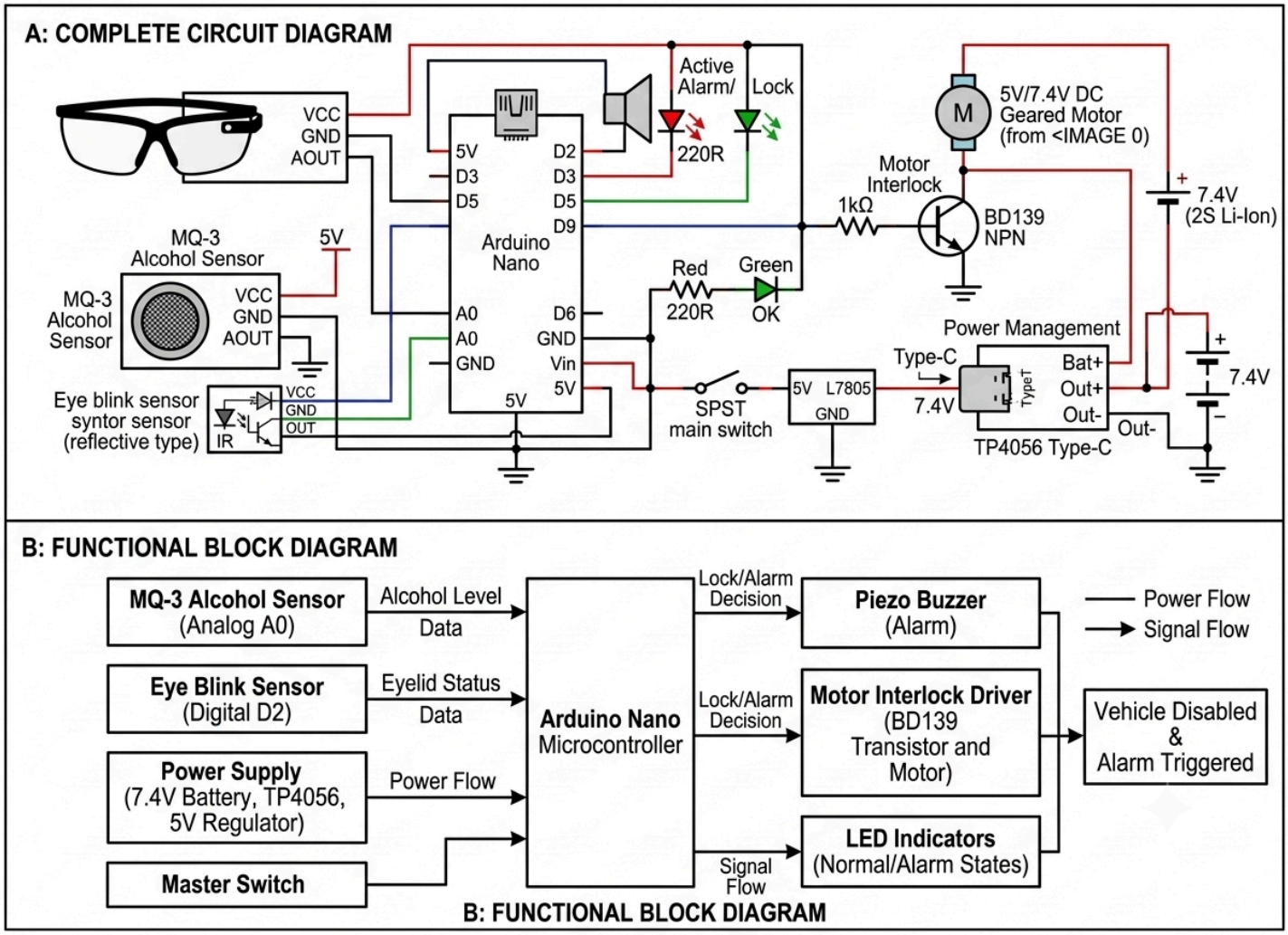
Purpose: The main purpose of this project is to improve road safety by detecting driver drowsiness and alcohol consumption.The system provides an alarm when the driver's eyes remain closed for approximately 3 seconds. If alcohol is detected, the system activates an alarm and prevents the motor from operating, representing an automatic engine-lock mechanism.

Methodology: The project uses an Arduino Nano as the main controller. An eye-blink sensor is used to check the driver's eyes. When the driver's eyes stay closed for 3 seconds, the Arduino turns ON the buzzer and LED to alert the driver.

An MQ-3 alcohol sensor is used to detect alcohol. When alcohol is detected, it sends a signal to the Arduino. The Arduino then turns ON the buzzer and stops the BO motor. The BO motor is used as a small model of the vehicle engine. A BD139 transistor is used to control the motor.

A 220 Ω resistor is connected with the LED, and a 1 kΩ resistor is used with the transistor. A switch is used to turn the system ON/OFF. The TP4056 Type-C module is used for charging the battery.

Block Diagram and Circuit Diagram:



Conclusion: The developed prototype successfully demonstrates an automated driver-safety system that monitors both eye closure and alcohol presence. It provides an audible and visual warning during drowsiness and stops the motor when an unsafe condition is detected.